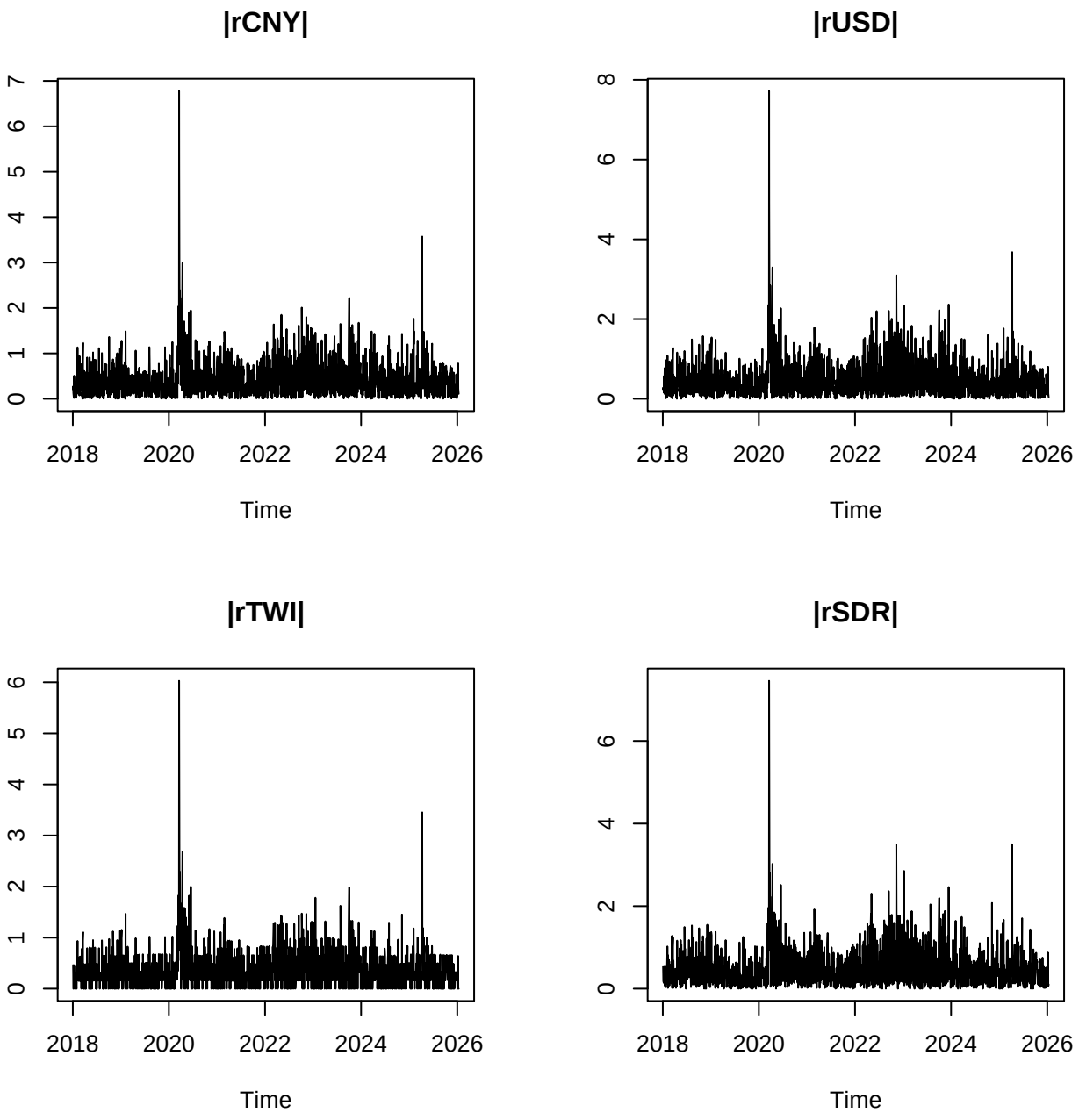
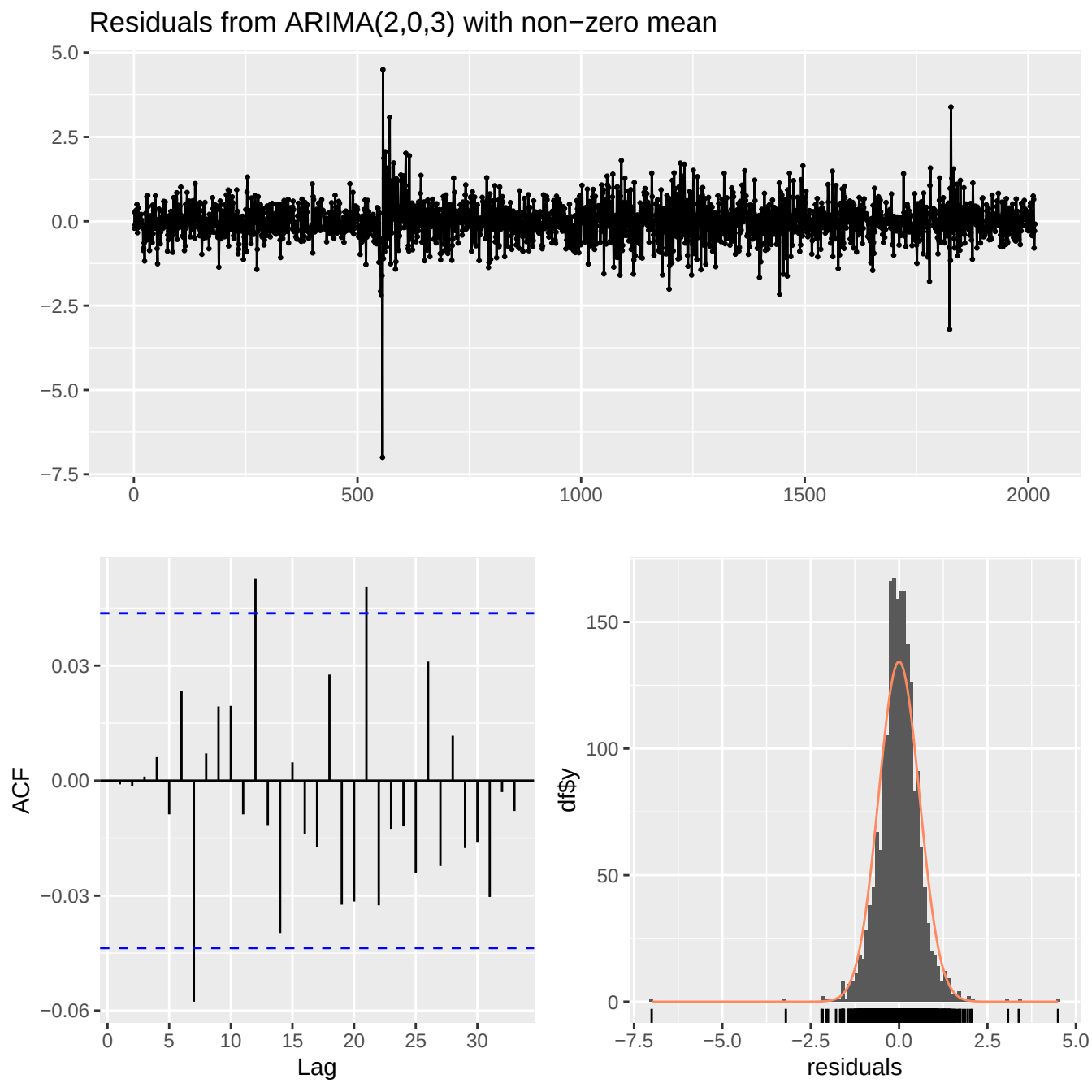
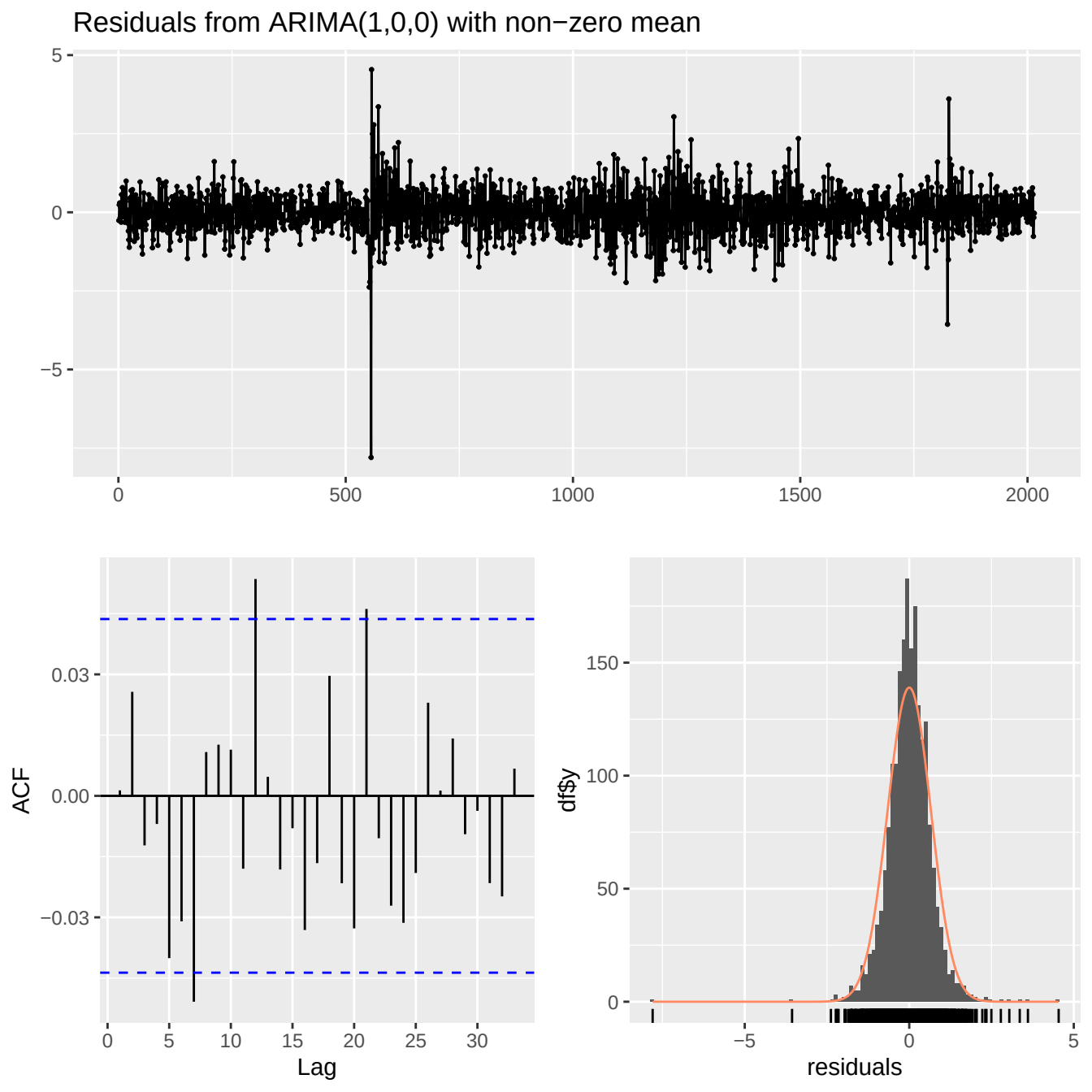
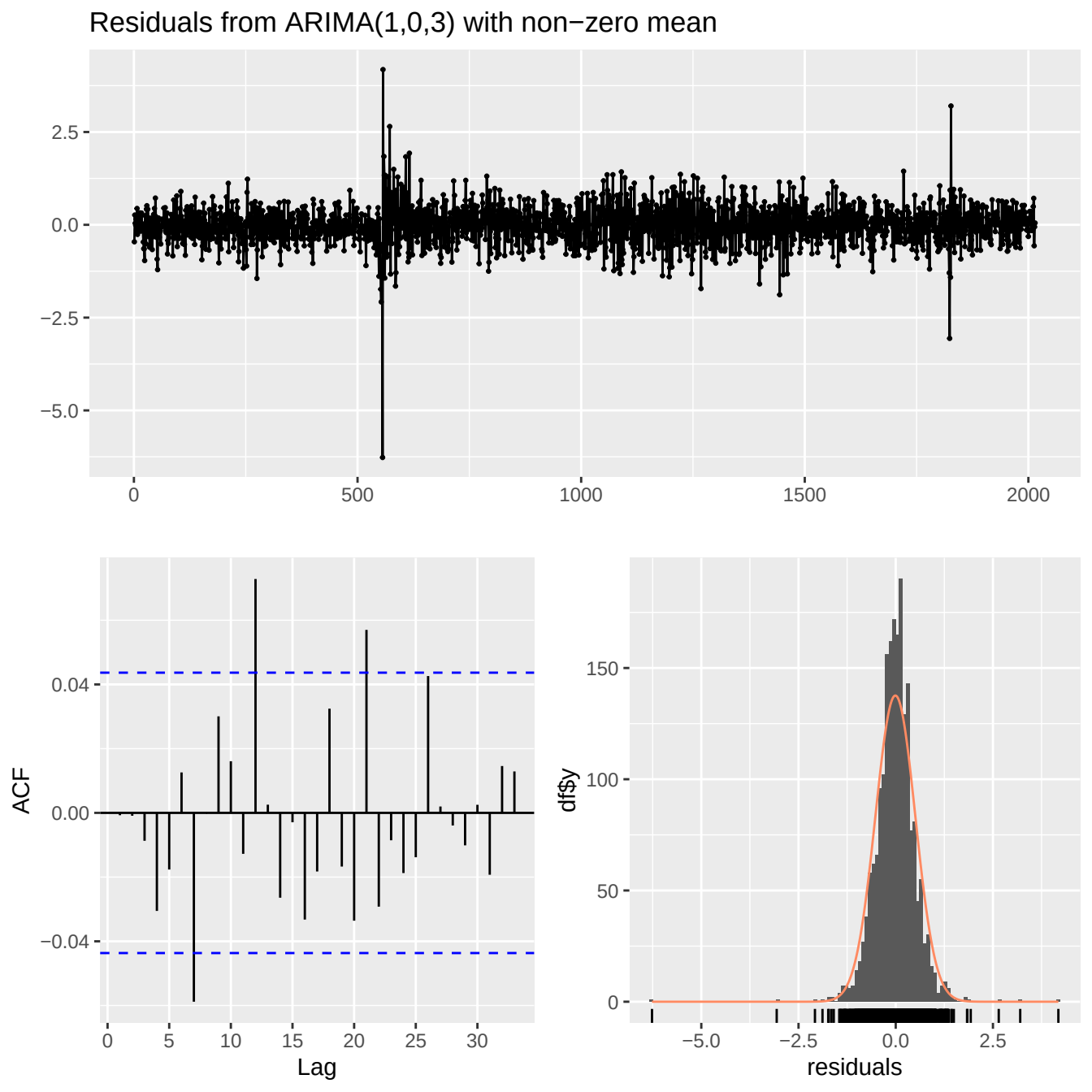
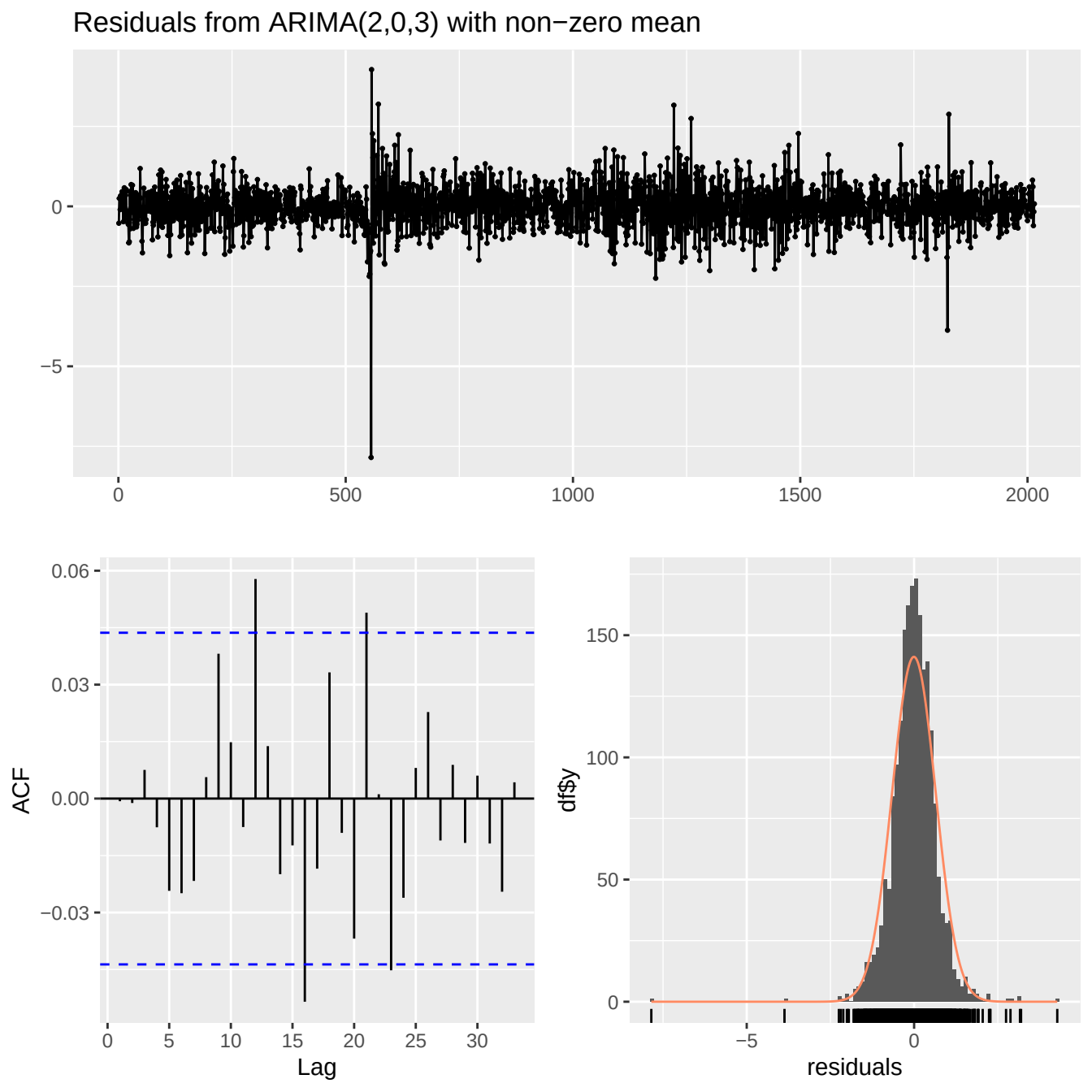
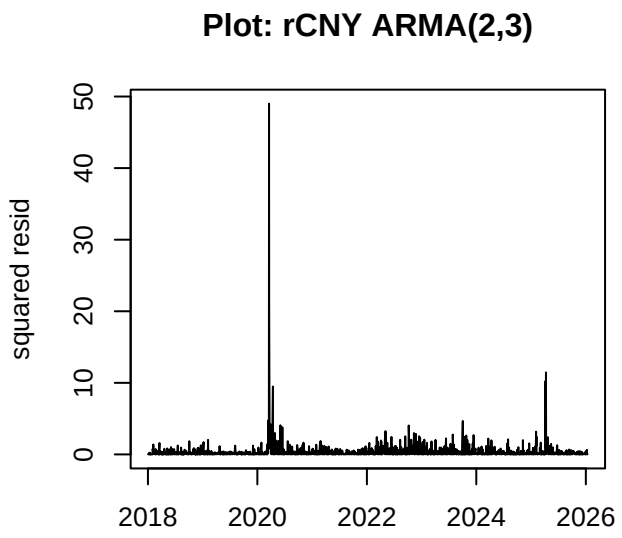

|rCNY|
0
1
2
3
4
5
6
7
2018
2020
2022
2024
2026
Time
|rUSD|
0
2
4
6
8
2018
2020
2022
2024
2026
Time
|rTWI|
0
1
2
3
4
5
6
2018
2020
2022
2024
2026
Time
|rSDR|
0
2
4
6
2018
2020
2022
2024
2026
Time

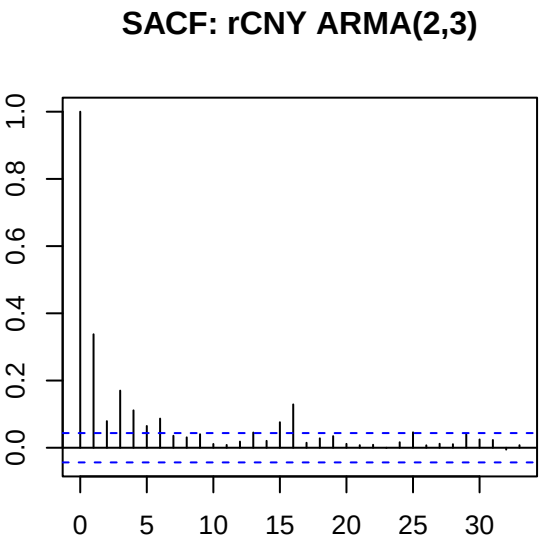

Residuals from ARIMA(2,0,3) with non−zero mean
5.0
2.5
0.0
−2.5
−5.0
−7.5
0
500
1000
1500
2000
ACF
0.03
0.00
−0.03
−0.06
0
5
10
15
20
25
30
Lag
df$y
150
100
50
0
−7.5
−5.0
−2.5
0.0
2.5
5.0
residuals

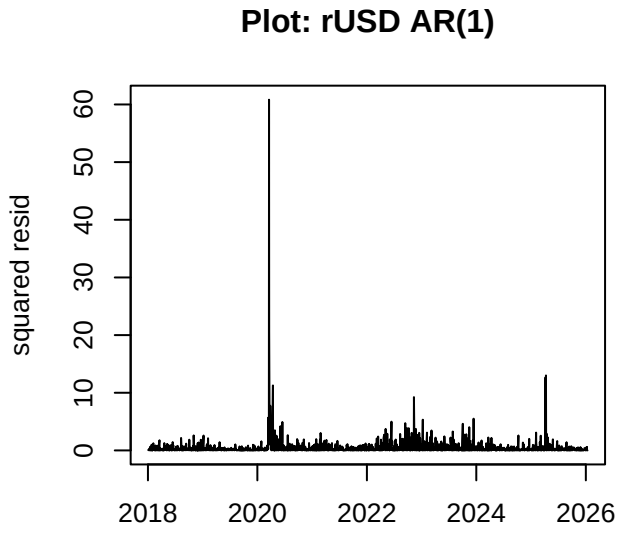

Residuals from ARIMA(1,0,0) with non−zero mean
5
0
−5
0
500
1000
1500
2000
ACF
0.03
0.00
−0.03
0
5
10
15
20
25
30
Lag
df$y
150
100
50
0
−5
0
5
residuals

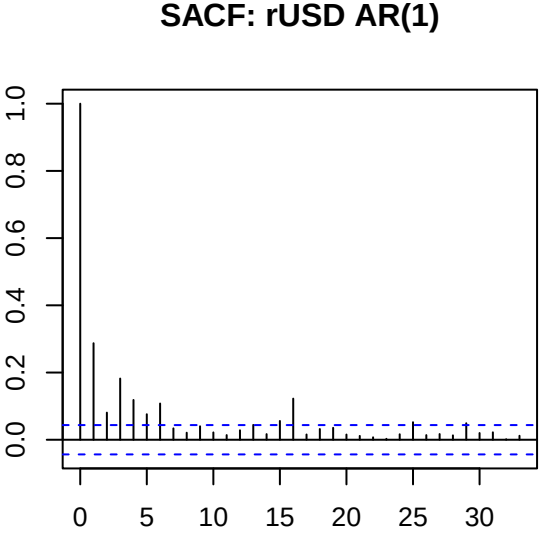

Residuals from ARIMA(1,0,3) with non−zero mean
2.5
0.0
−2.5
−5.0
0
500
1000
1500
2000
ACF
0.04
0.00
−0.04
0
5
10
15
20
25
30
Lag
df$y
150
100
50
0
−5.0
−2.5
0.0
2.5
residuals

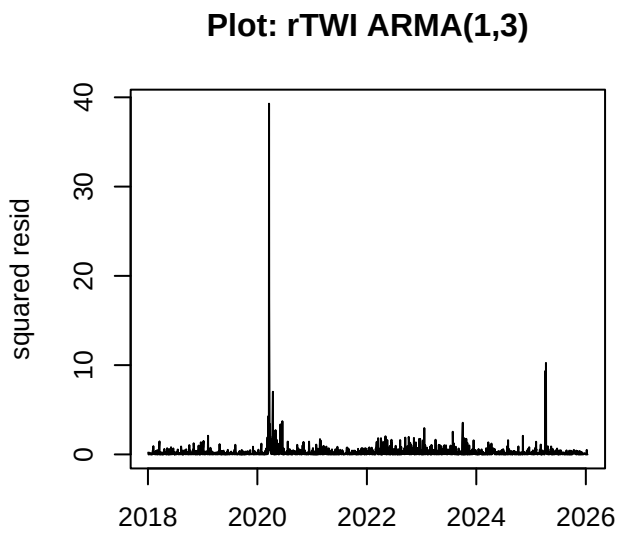

Residuals from ARIMA(2,0,3) with non−zero mean
0
−5
0
500
1000
1500
2000
0.06
0.03
0.00
−0.03
ACF
0
5
10
15
20
25
30
Lag
150
100
50
0
df$y
−5
0
residuals

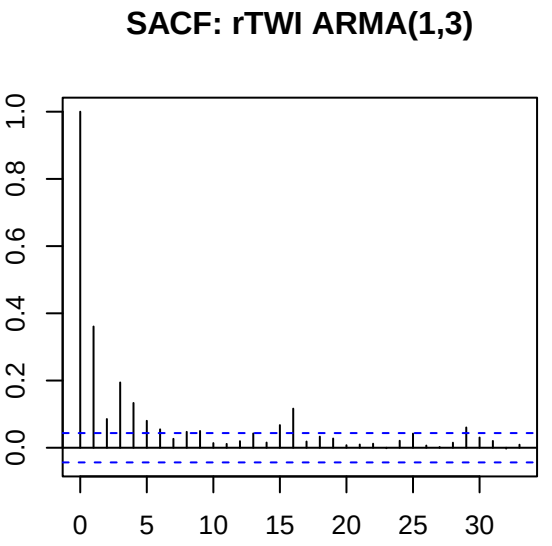

Plot: rCNY ARMA(2,3)
squared resid
0
10
20
30
40
50
2018
2020
2022
2024
2026

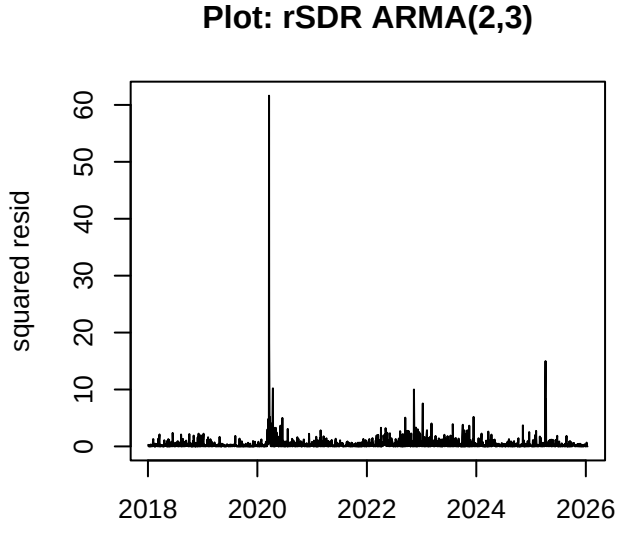

SACF: rCNY ARMA(2,3)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30

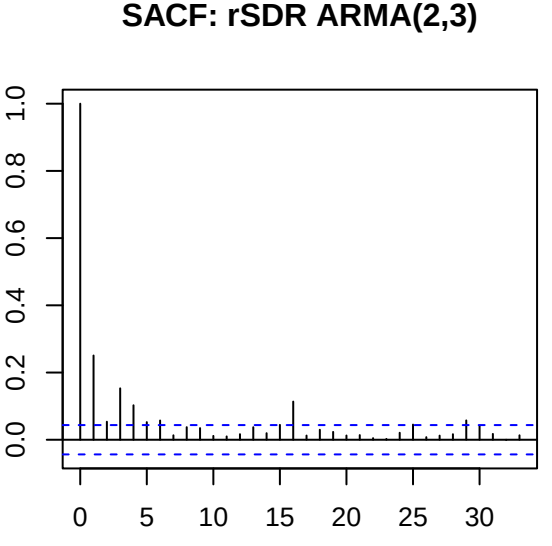

Plot: rUSD AR(1)
squared resid
0
10
20
30
40
50
60
2018
2020
2022
2024
2026

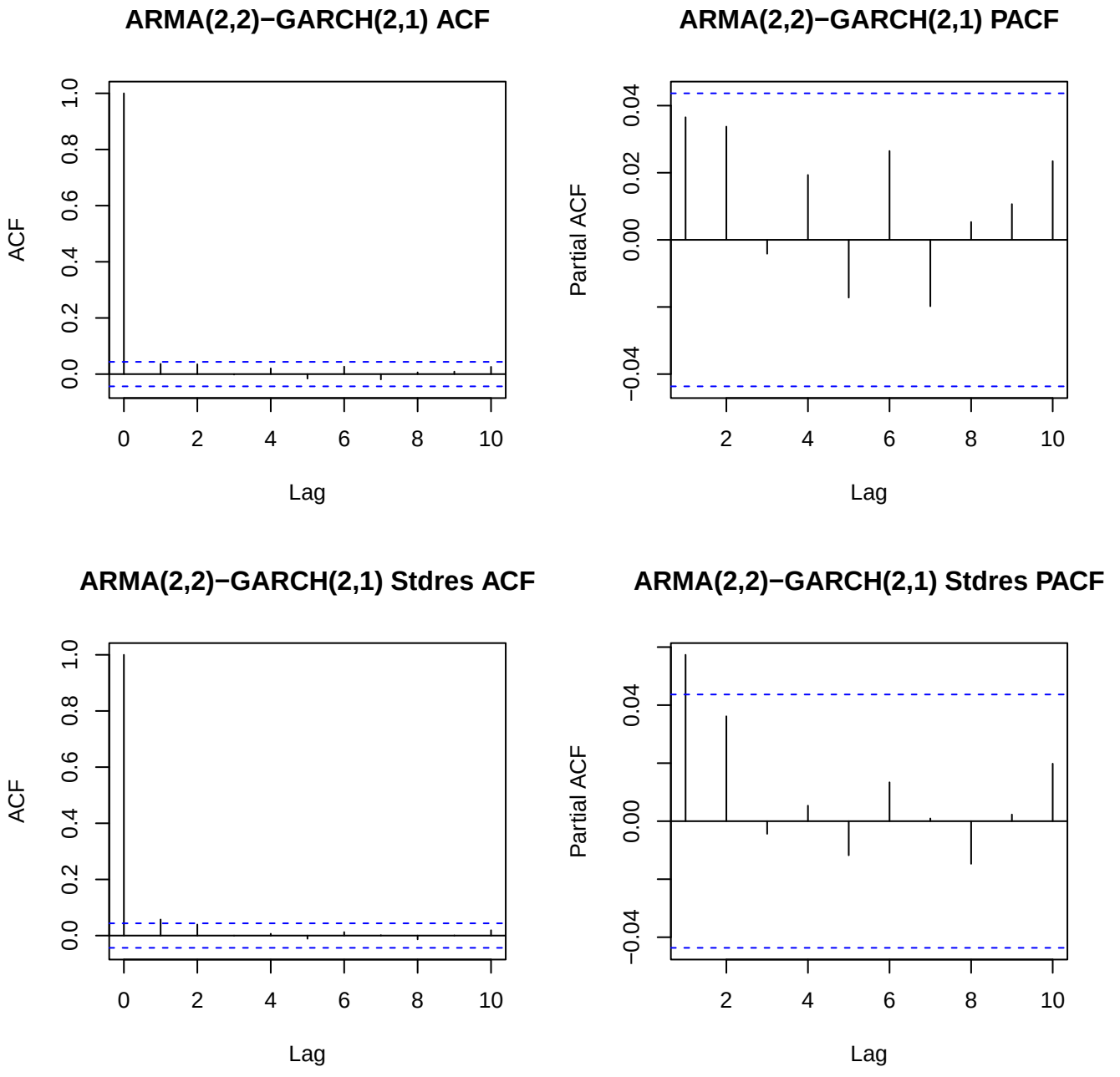

SACF: rUSD AR(1)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30

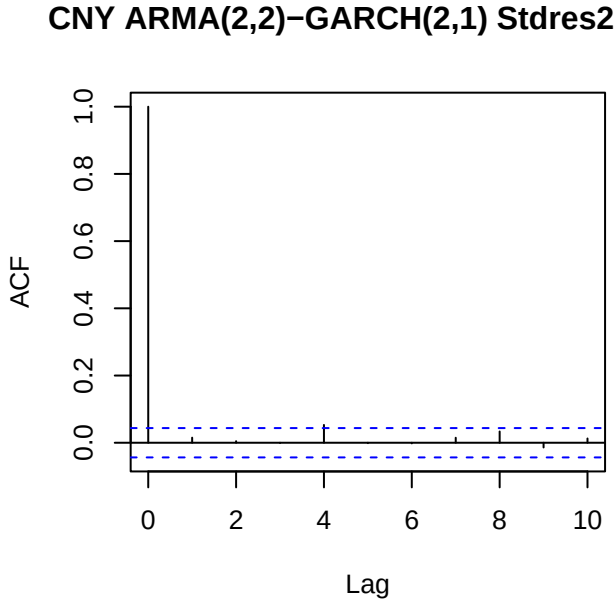

Plot: rTWI ARMA(1,3)
squared resid
0
10
20
30
40
2018
2020
2022
2024
2026

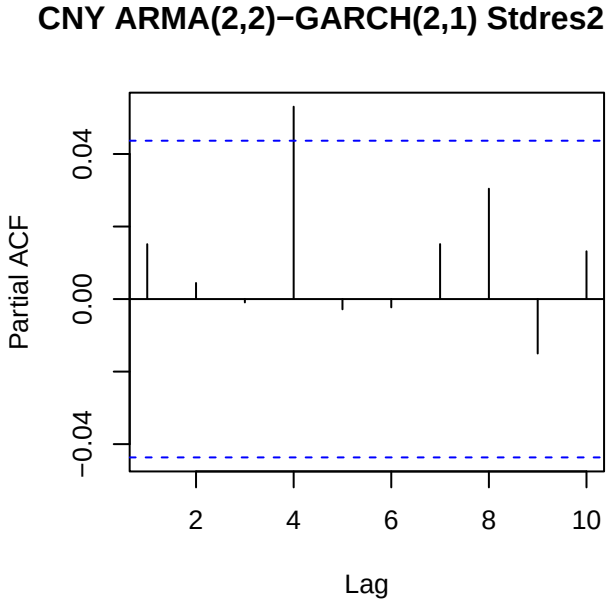

SACF: rTWI ARMA(1,3)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30

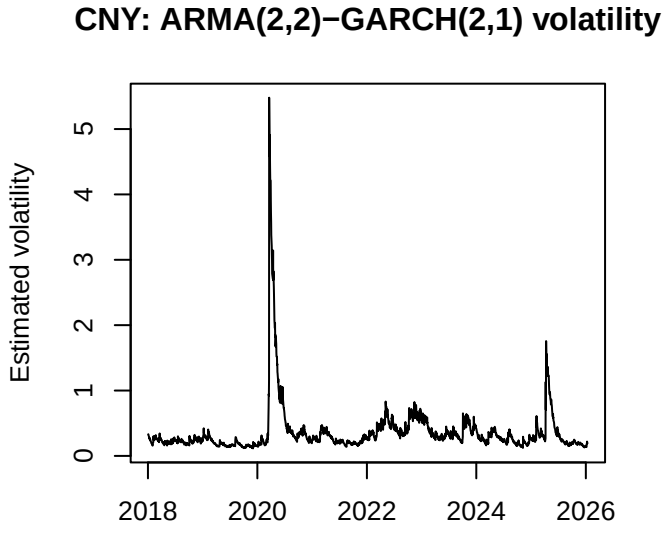

Plot: rSDR ARMA(2,3)
squared resid
0
10
20
30
40
50
60
2018
2020
2022
2024
2026

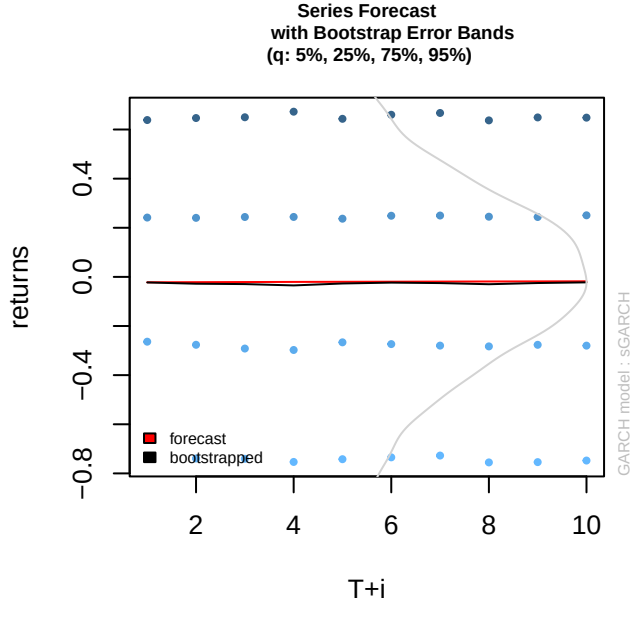

SACF: rSDR ARMA(2,3)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30

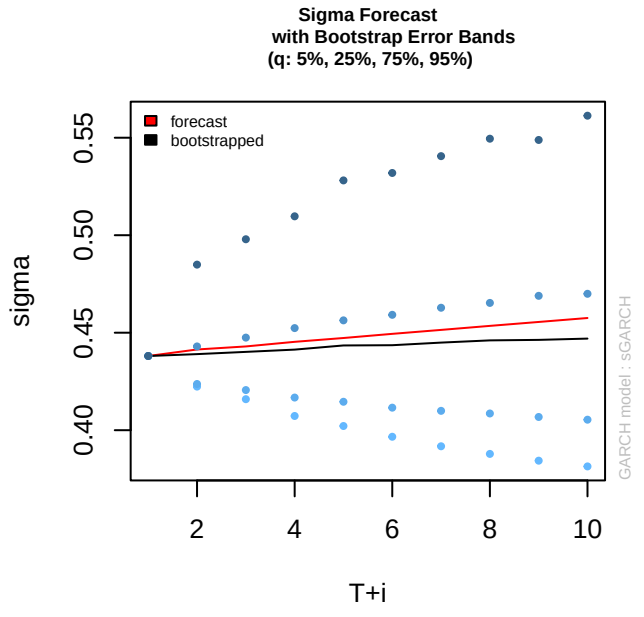

ARMA(2,2)−GARCH(2,1) ACF
ACF
Lag
ARMA(2,2)−GARCH(2,1) PACF
Partial ACF
Lag
ARMA(2,2)−GARCH(2,1) Stdres ACF
ACF
Lag
ARMA(2,2)−GARCH(2,1) Stdres PACF
Partial ACF
Lag

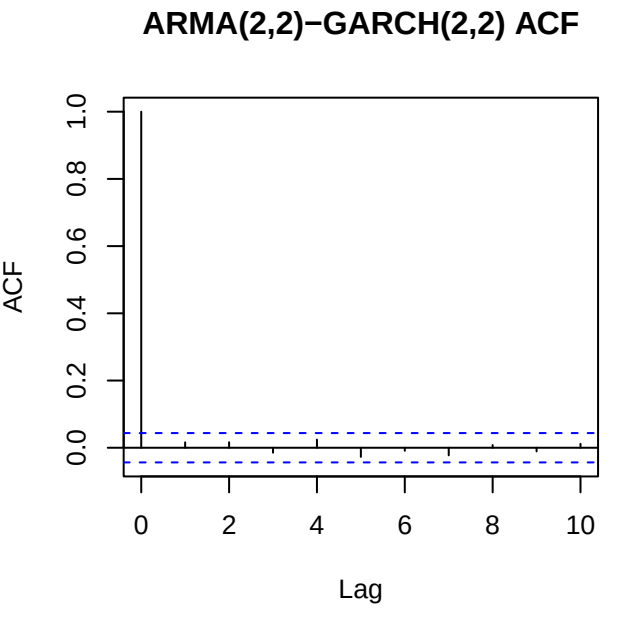

CNY ARMA(2,2)−GARCH(2,1) Stdres2 ACF
ACF
Lag

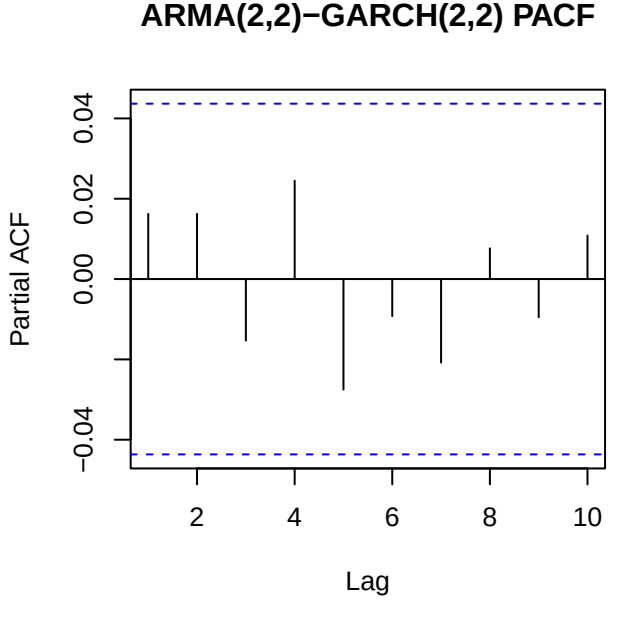

CNY ARMA(2,2)−GARCH(2,1) Stdres2 PACF
Partial ACF
Lag

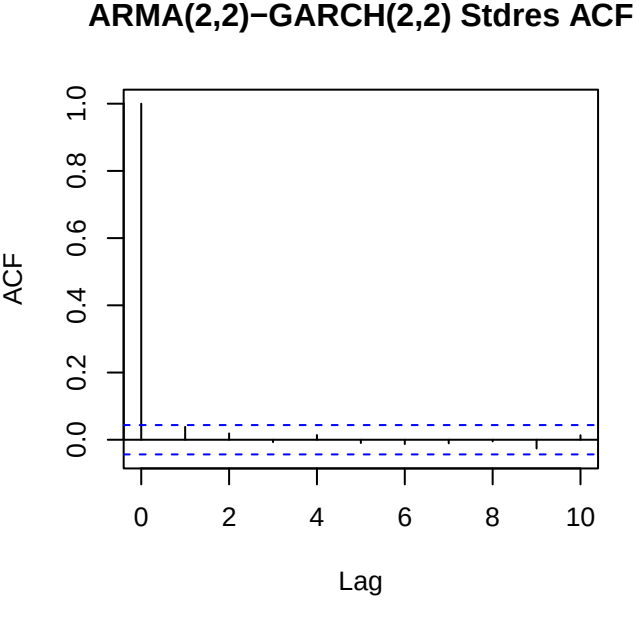

CNY: ARMA(2,2)−GARCH(2,1) volatility
Estimated volatility

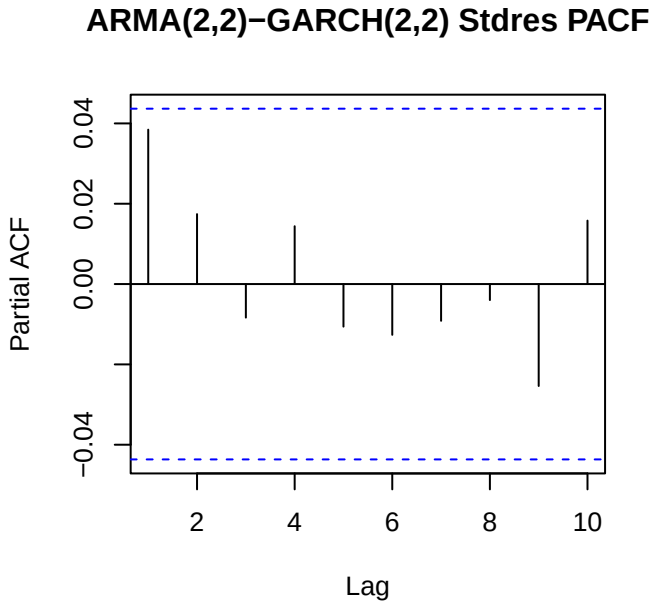

Series Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
returns
T+i
forecast
bootstrapped
GARCH model : sGARCH

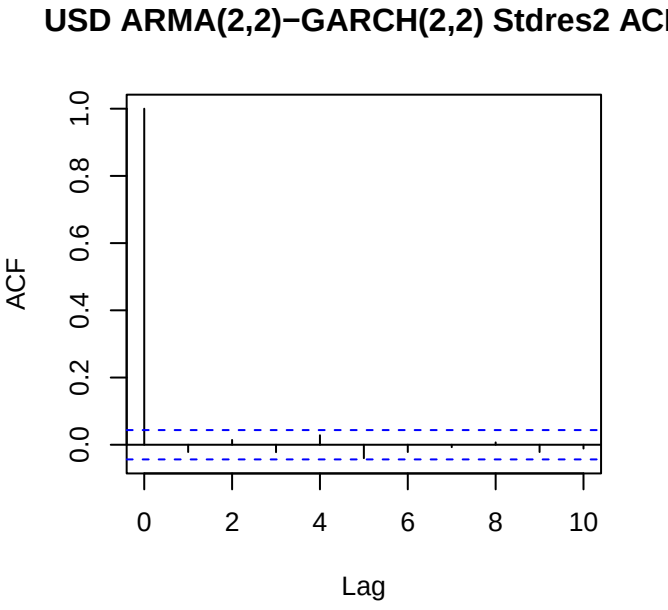

Sigma Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
forecast
bootstrapped
sigma
0.40
0.45
0.50
0.55
2
4
6
8
10
T+i
GARCH model : sGARCH

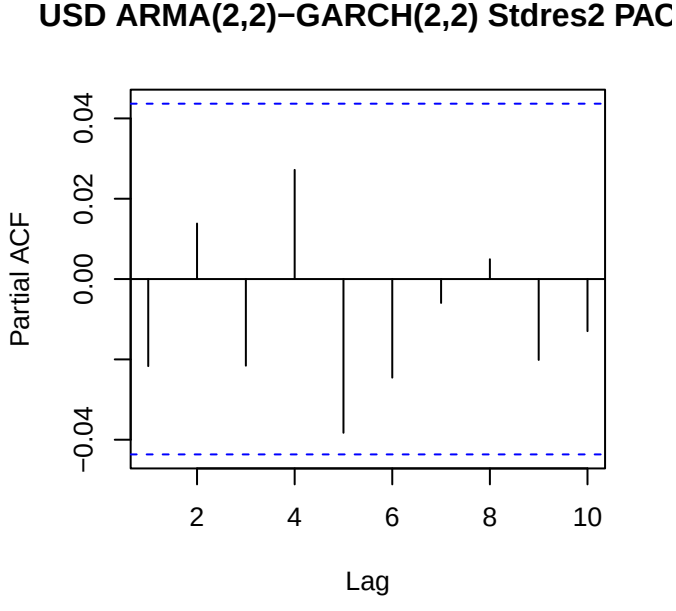

ARMA(2,2)−GARCH(2,2) ACF
ACF
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
Lag

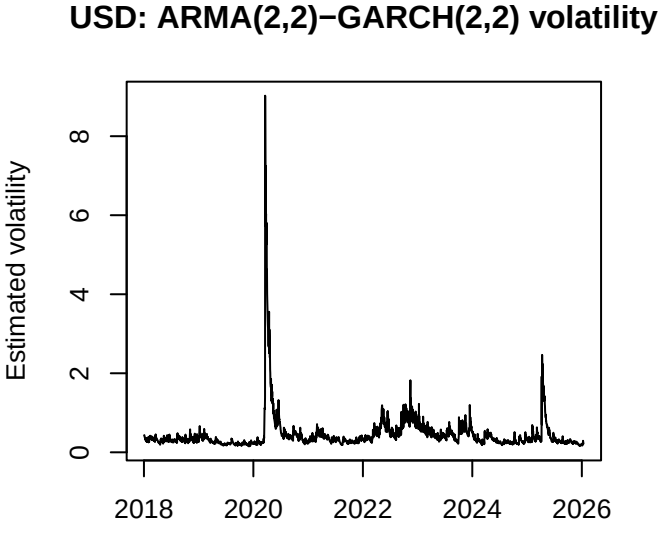

ARMA(2,2)−GARCH(2,2) PACF
Partial ACF
−0.04
0.00
0.02
0.04
2
4
6
8
10
Lag

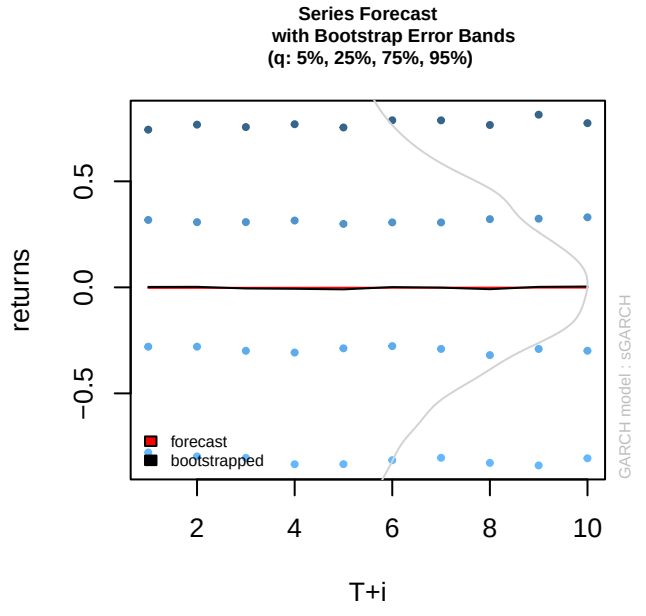

ARMA(2,2)−GARCH(2,2) Stdres ACF
ACF
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
Lag

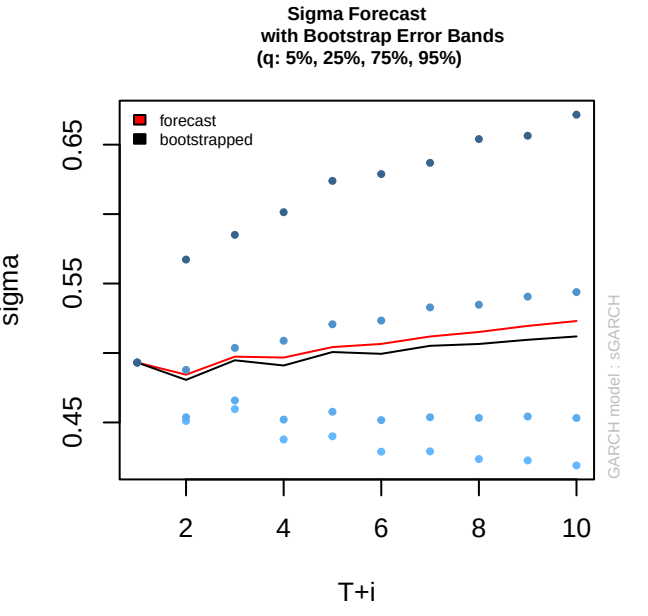

ARMA(2,2)−GARCH(2,2) Stdres PACF
Partial ACF
0.04
0.02
0.00
−0.04
2
4
6
8
10
Lag

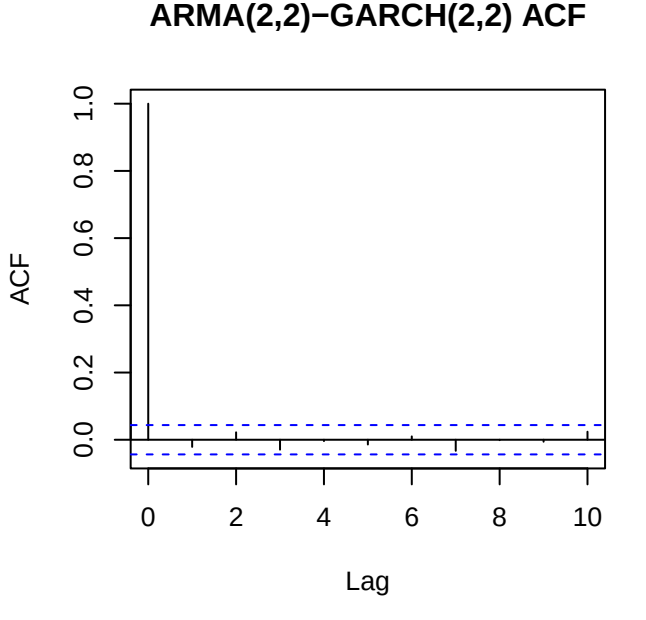

USD ARMA(2,2)−GARCH(2,2) Stdres2 ACF
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
Lag

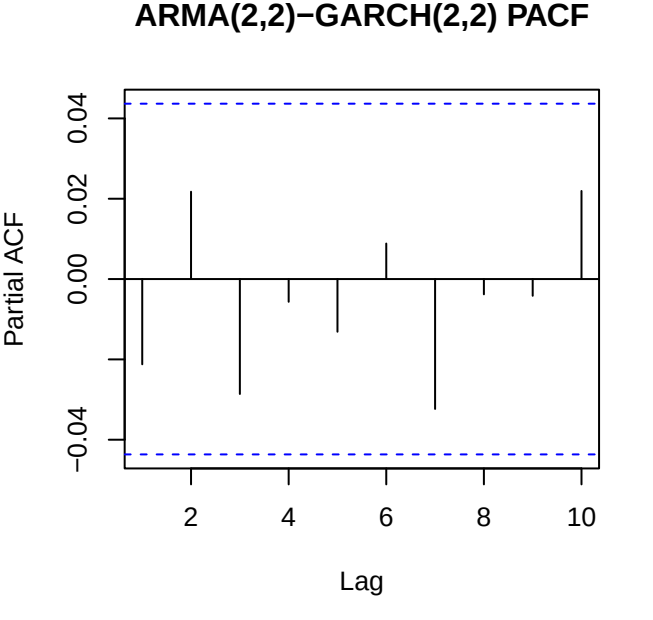

USD ARMA(2,2)−GARCH(2,2) Stdres2 PAC
Partial ACF
0.04
0.02
0.00
−0.04
2
4
6
8
10
Lag

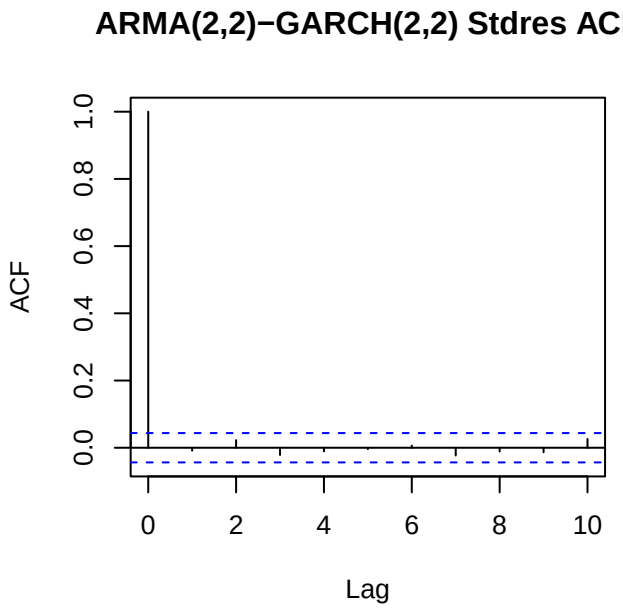

USD: ARMA(2,2)−GARCH(2,2) volatility
Estimated volatility
8
6
4
2
0
2018
2020
2022
2024
2026

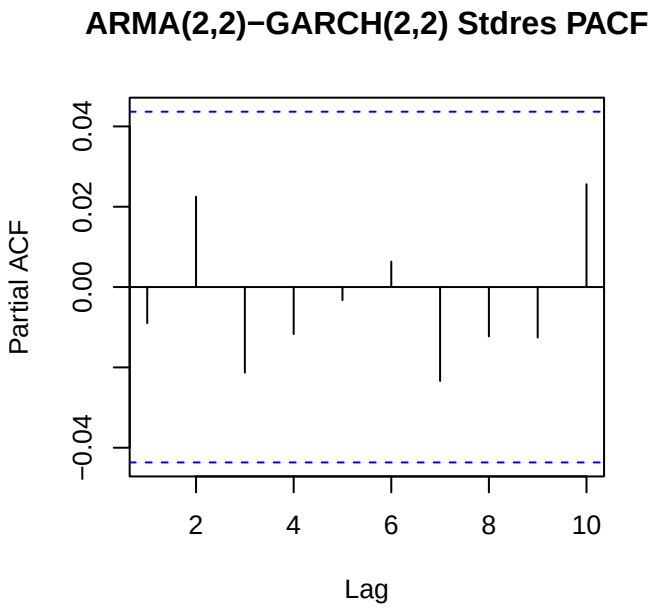

Series Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
returns
T+i
forecast
bootstrapped
GARCH model : sGARCH

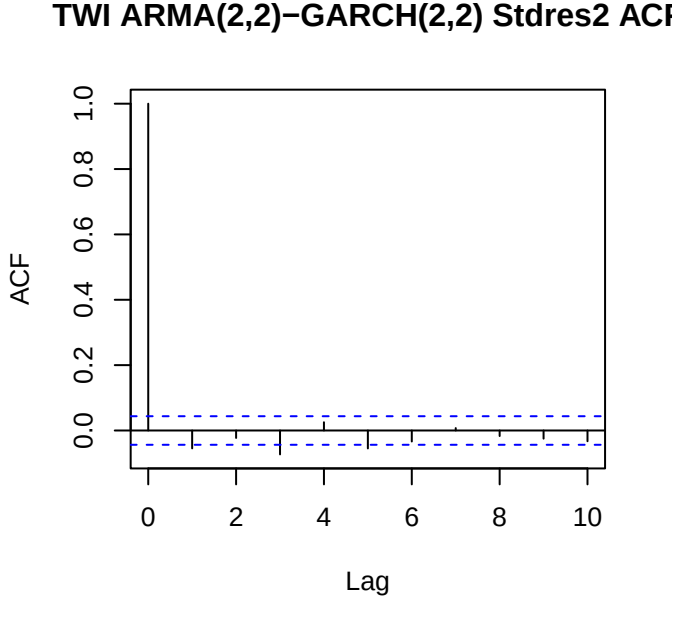

Sigma Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
sigma
T+i
forecast
bootstrapped
GARCH model : sGARCH

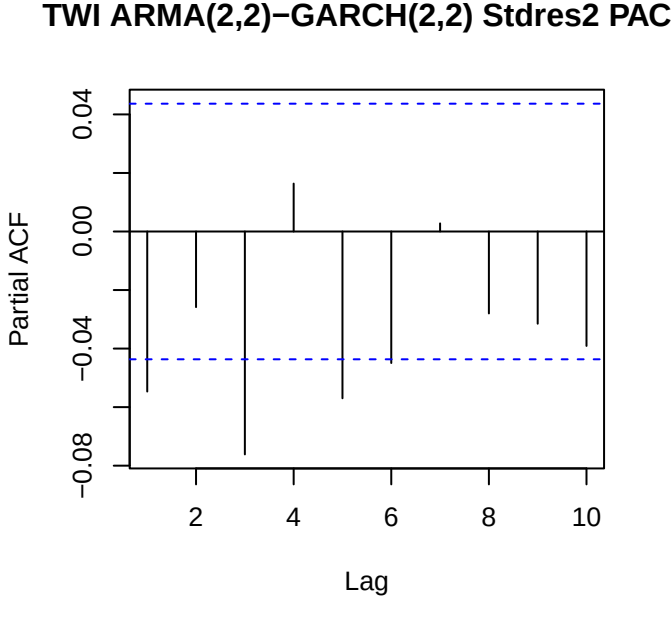

ARMA(2,2)−GARCH(2,2) ACF
ACF
Lag

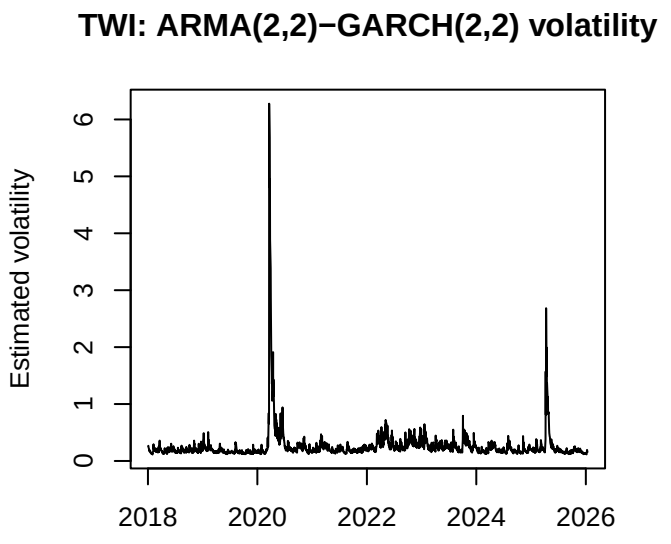

ARMA(2,2)−GARCH(2,2) PACF
Partial ACF
Lag

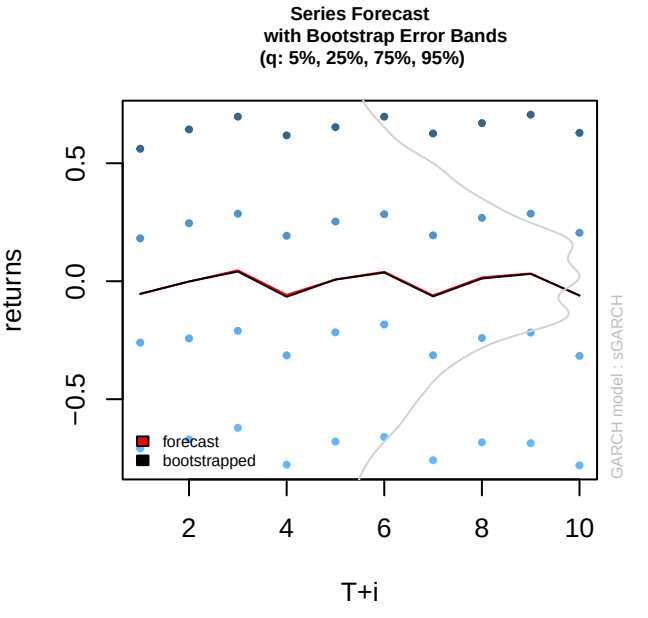

ARMA(2,2)−GARCH(2,2) Stdres ACF
ACF
Lag

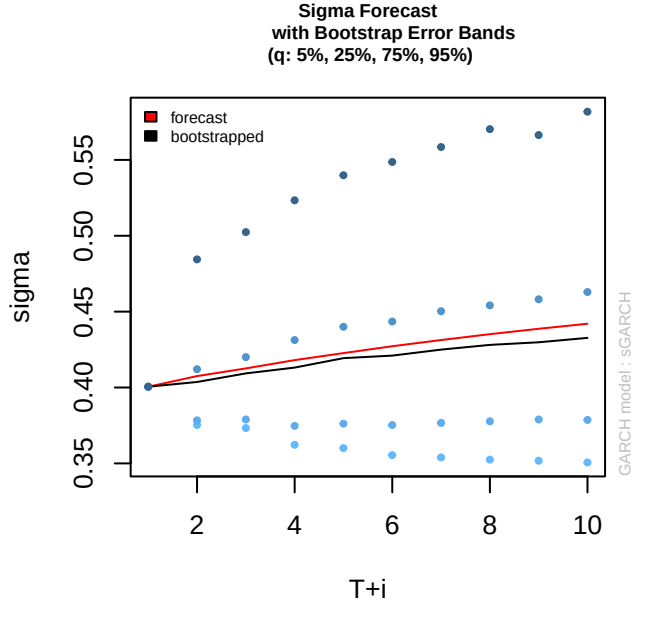

ARMA(2,2)−GARCH(2,2) Stdres PACF
Partial ACF
Lag

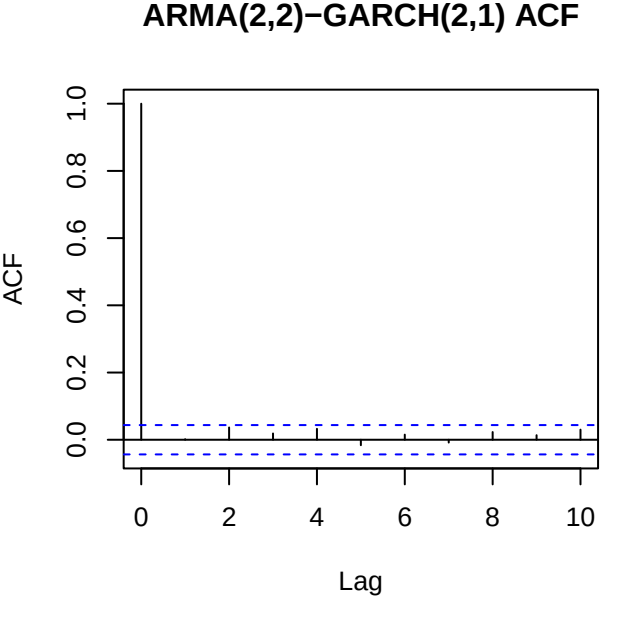

TWI ARMA(2,2)−GARCH(2,2) Stdres2 ACI
ACF
Lag

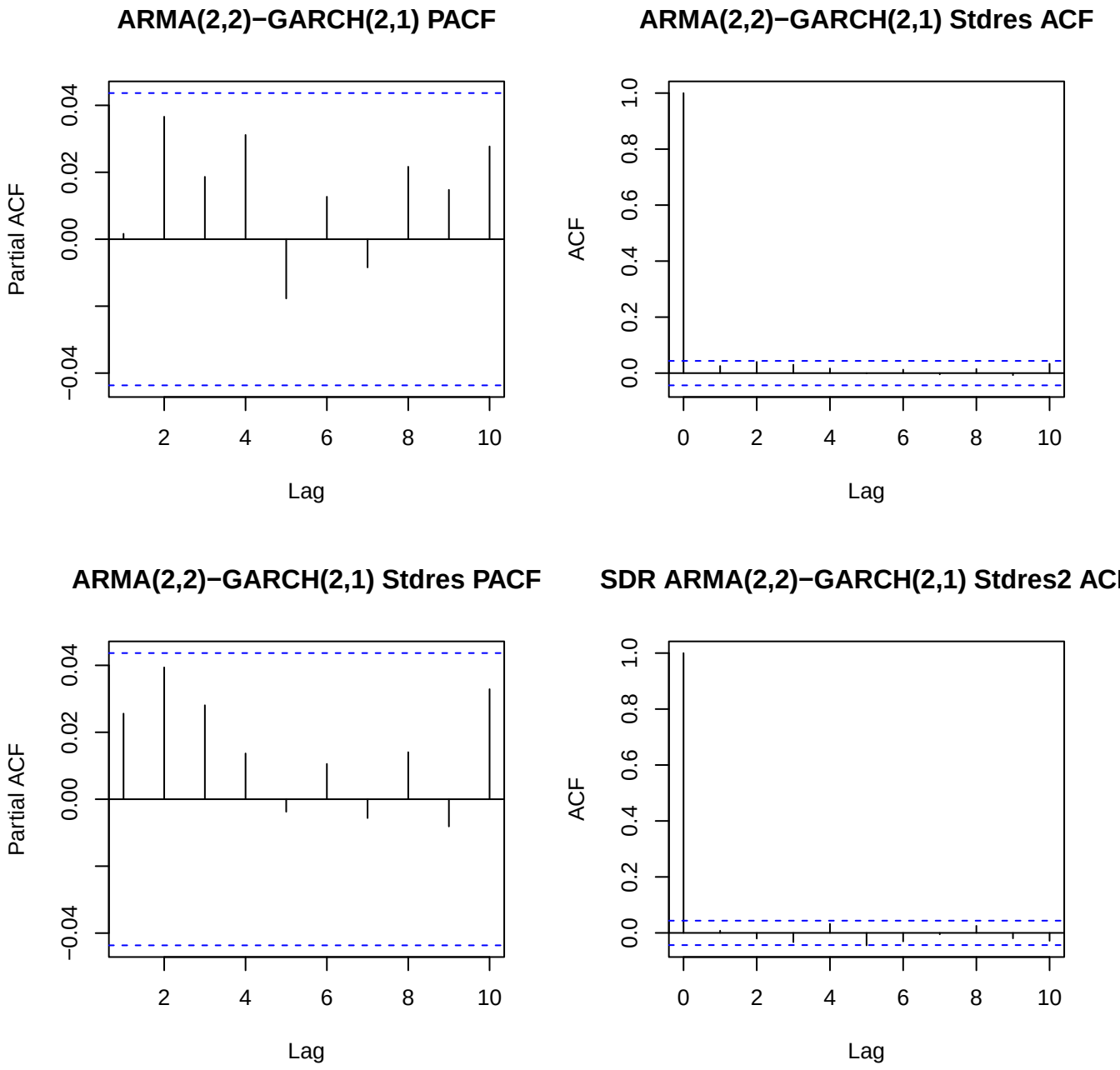

TWI ARMA(2,2)−GARCH(2,2) Stdres2 PAC
Partial ACF
Lag

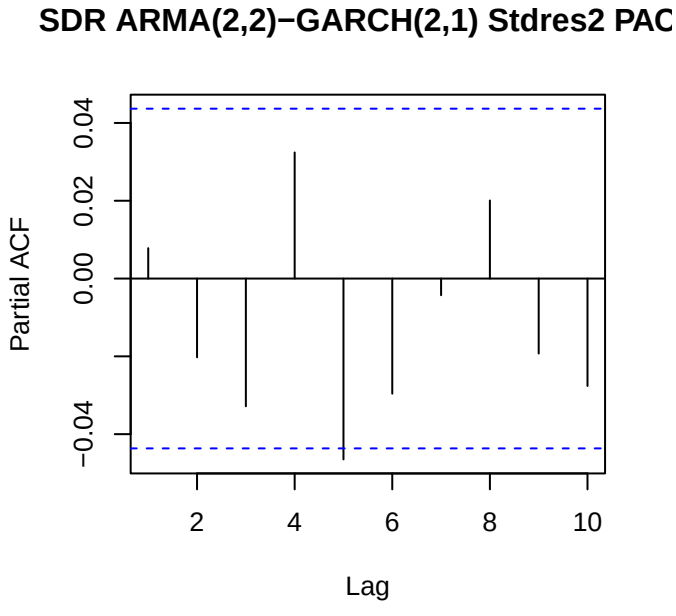

TWI: ARMA(2,2)−GARCH(2,2) volatility
Estimated volatility
0
1
2
3
4
5
6
2018
2020
2022
2024
2026

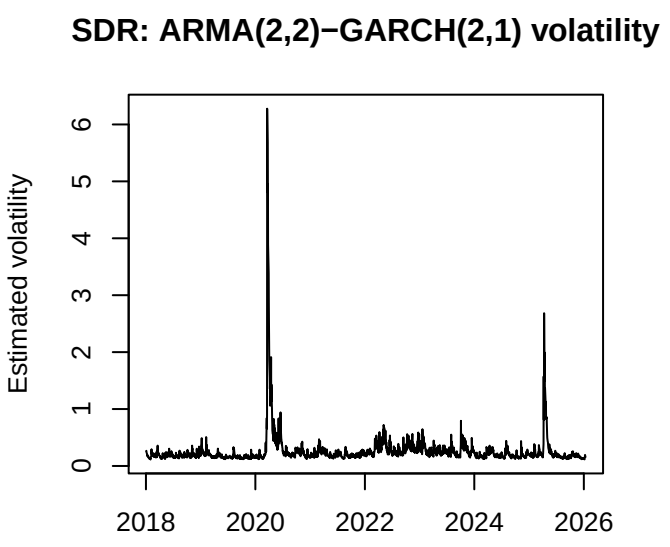

Series Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
returns
−0.5
0.0
0.5
forecast
bootstrapped
2
4
6
8
10
T+i
GARCH model : sGARCH

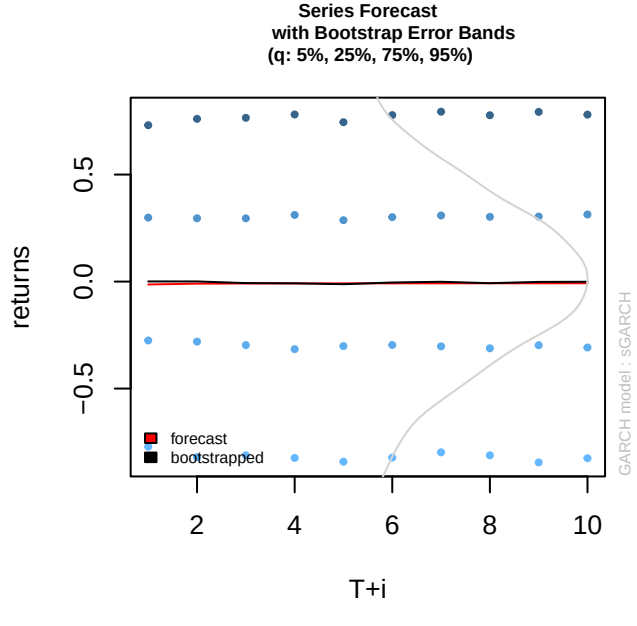

Sigma Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
sigma
0.35
0.40
0.45
0.50
0.55
forecast
bootstrapped
2
4
6
8
10
T+i
GARCH model : sGARCH

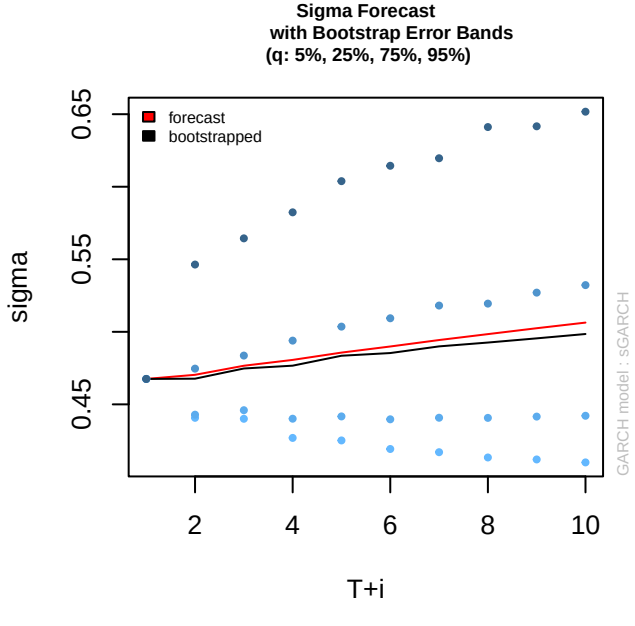

ARMA(2,2)−GARCH(2,1) ACF
ACF
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
Lag


ARMA(2,2)−GARCH(2,1) PACF
Partial ACF
Lag
ARMA(2,2)−GARCH(2,1) Stdres ACF
ACF
Lag
ARMA(2,2)−GARCH(2,1) Stdres PACF
Partial ACF
Lag
SDR ARMA(2,2)−GARCH(2,1) Stdres2 ACF
ACF
Lag


SDR ARMA(2,2)−GARCH(2,1) Stdres2 PAC
Partial ACF
0.04
0.02
0.00
−0.04
2
4
6
8
10
Lag


SDR: ARMA(2,2)−GARCH(2,1) volatility
Estimated volatility
0
1
2
3
4
5
6
2018
2020
2022
2024
2026


Series Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
returns
0.5
0.0
−0.5
forecast
bootstrapped
GARCH model : sGARCH
2
4
6
8
10
T+i


Sigma Forecast
with Bootstrap Error Bands
(q: 5%, 25%, 75%, 95%)
sigma
0.65
0.55
0.45
forecast
bootstrapped
GARCH model : sGARCH
2
4
6
8
10
T+i